MH_TechHydrMotion.library
MH_ForceLimitOn

Brief description

The function block enables the alternating control of the axis and specifies a (target) force command value. The command value generation is defined via a ramp or a time.

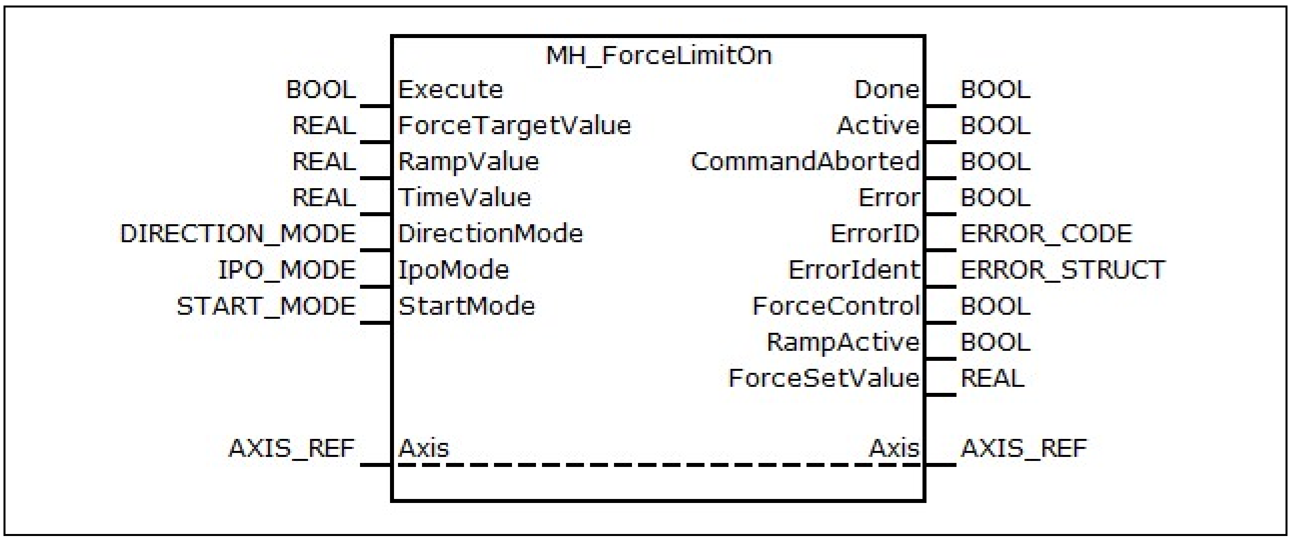
The force command value is specified by the force controller of the axis.

Assignment: Target system/library

Target system	Library
IndraMotion MLC from version 14VRS HYD basic package	MH_TechHydrMotion

Reference table of the MH_ForceLimitOn

Interface description



MH_ForceLimitOn function block

	Name	Type	Comment
VAR_INPUT	Execute	BOOL	Function block processing enabled
	ForceTargetValue	REAL	New command force to which it is to be controlled and blended should be acc. to unit/scaling of the axis
	RampValue	REAL	Ramp to be used for blending. Force unit per second
	TimeValue	REAL	Time in seconds while blending has to take place
	DirectionMode	DIRECTION_MODE	POSITIVE = Effective in positive direction of force NEGATIVE = Effective in negative

			direction of force BIPOLAR = Effective in both directions of force
	IpoMode	IPO_MODE	Interpolation mode: CONT (continue), BREAK, JUMP
	StartMode	START_MODE	NOW = Command value generation starts immediately IN_FORCE_CONTROL = The command value generation starts when the force controller is active
VAR_OUTPUT	Done	BOOL	Processing completed. The outputs are valid
	Active	BOOL	The function block operates. The outputs are valid
	CommandAborted	BOOL	The function block was aborted by another "Force" function block
	Error	BOOL	Indicates that an error occurred in the function block
	ErrorID	ERROR_CODE	Error ID
	ErrorIdent	ERROR_STRUCT	Error structure with further error division, see Error codes
	ForceControl	BOOL	Force controller of axis effective
	RampActive	BOOL	Ramp generation running
	ForceSetValue	REAL	Force command value currently sent to the force controller of the axis
VAR_IN_OUT CONSTANT	Axis	AXIS_REF	Contains information on the specific axis
PROPERTY	CycleTime	REAL	Cycle time of the function block in µs. If "Property" is not set, the cycle time is automatically determined while enabling. In this case, the processing is delayed by one cycle.

MH_ForceLimitOn function block

Name	Type	Min. value	Max. value	Default value	Effective
Execute	BOOL			FALSE	Continuous
ForceTargetValue	REAL	n. def.	n. def.	0.0	Rising edge at "Execute"
RampValue	REAL	0.0	n. def.	0.0	Rising edge at "Execute"

TimeValue	REAL	0.0	n. def.	0.0	Rising edge at “Execute”
DirectionMode	DIRECTION_MODE			POSITIVE	Rising edge at “Execute”
IpoMode	IPO_MODE			CONT	Rising edge at “Execute”
StartMode	START_MODE			NOW	Rising edge at “Execute”

Input behavior of the MH_ForceLimitOn

Functional description

At a rising edge at the "Execute" input, the initialization phase is run through and the force limitation (alternating control) is enabled.

The starting force command value depends on whether the alternating control has already be enabled before activating the command. If the alternating command is not active before starting the function block, the actual force value is used as starting force command value. If the alternating control is already active, the latest written force command value is used as starting value.

Depending on the StartMode, the command value generation either starts immediately (NOW) or when the force controller is active (IN_FORCE_CONTROL). The parameters of the alternating control are used to set when the force controller is active.

If null is to be added to the inputs “RampValue” and “TimeValue”, the force command value of the axis is set to the target value with a jump. The ramp can be defined via a slope (RampValue) or a time specification (TimeValue).

Call the function block in the Motion task to ensure a synchronous command value generation.

When using the function block, ensure a previous initialization of the function block MH_InitHydr and the cyclic call of the function block MH_ForceLimitBase. The MH_InitHydr function block is provided by the MH_TechHydrBase library.

Error codes

The function block uses the error table **F_RELATED_TABLE**, 16#0170.

It can generate the following error messages in Additional1/Additional2:

ErrorID	**Additional1**	**Additional2**	**Description**
ACCESS_ERROR	16#00002410	16#00000000	Access to an unsupported axis. The HydraulicDrive and the hydraulic controller axis are supported
ACCESS_ERROR	16#00002411	16#00000000	MH_HydrInit not available/successfully started
RESOURCE_ERROR	16#00002417	16#00000000	MH_ForceLimitBase not available/successfully started

Error numbers of the MH_ForceLimitOn function block

 RS-50eb2e25daeb518a0a347ea500182240-3-en-US-2
DOK-MLC***-TF*HMOT*V14-LI05-EN-P